



Caching Content

AWS Academy Cloud
Architecting



Introduction

Caching Content

Module objectives

This module prepares you to do the following:

- Identify how caching content can improve application performance and reduce latency.
- Identify how to use Amazon CloudFront to deliver content by using edge locations protection.
- Create architectures that use CloudFront to cache content.
- Describe how to use Amazon ElastiCache for database caching.
- Use the AWS Well-Architected Framework principles when designing caching strategies.

Module overview

Presentation sections

- Overview of caching
- Caching using CloudFront
- Caching using ElastiCache
- Applying AWS Well-Architected Framework principles to caching

Knowledge checks

- 10-question knowledge check
- Sample exam question

Hands-on labs in this module

Guided lab

- Streaming Dynamic Content Using Amazon CloudFront

As a cloud architect designing content caching:



- I need to determine when and how to provide a faster way for my customers to access content so that I can balance performance, cost, and data freshness to meet evolving business requirements.
- I need to incorporate a content delivery network (CDN) service in my architecture designs so that I can distribute static content securely and efficiently.
- I need to select an in-memory caching strategy so that I can reduce network latency and offload database pressure.



Overview of caching

Caching Content

Why cache content?

A cache is a high-speed data storage layer that stores a subset of data so that future requests for that data are served up faster.



Store regularly accessed data in a more optimized way.

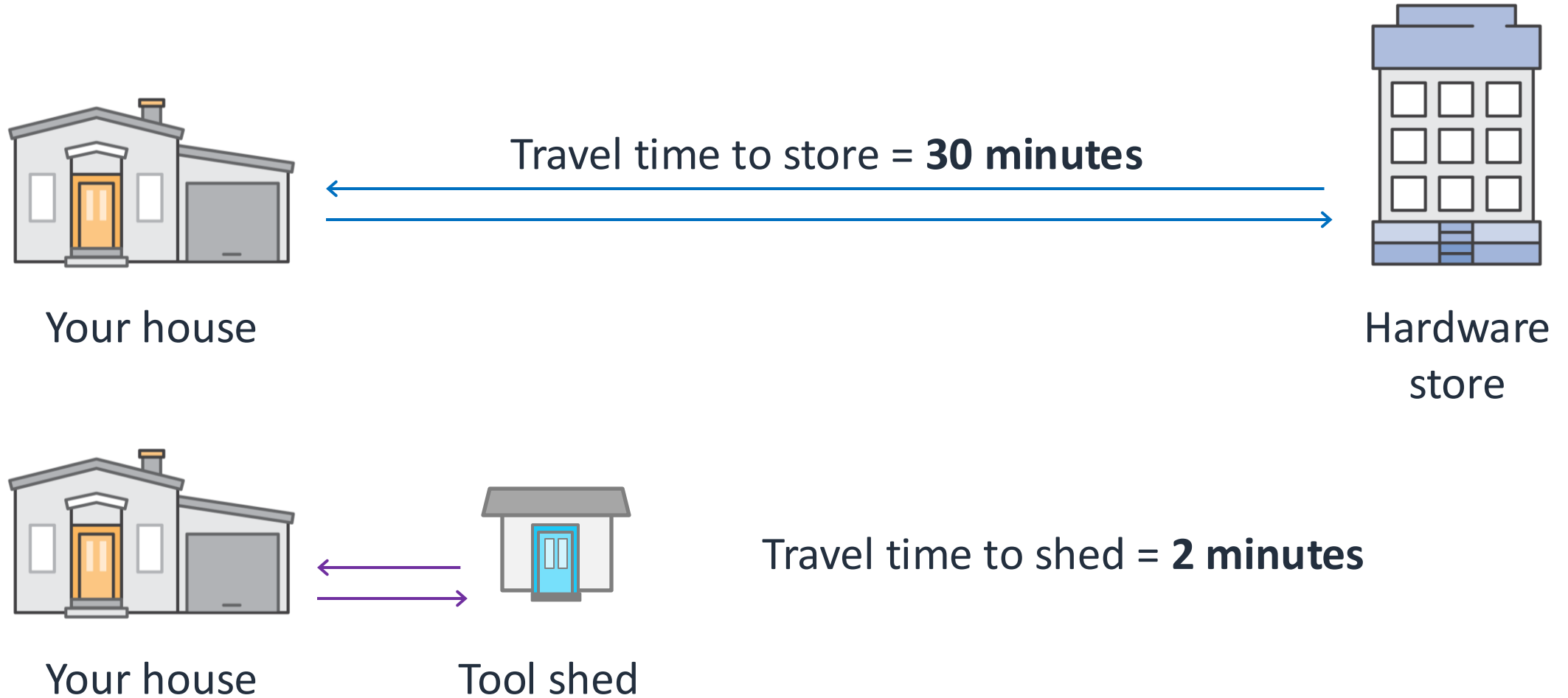


Increase data retrieval performance.



Reduce the need to access the underlying slower storage layer.

Caching analogy



What should you cache?



Static and frequently
accessed data



Results of
computationally
intensive calculations



Results of time-consuming,
frequently used, or
complex database queries

Trade-offs to consider when caching

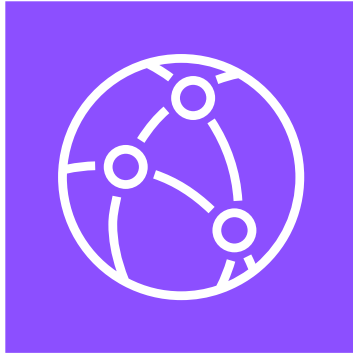
Benefits of caching

- Reduces response latency
- Reduces cost
- Alleviates the load on the origin (data source)
- Provides predictable performance
- Improves availability

Challenges of caching

- Requires additional engineering
- Requires someone to determine how to cache data and deal with stale data based on the business use case

Types of caching and AWS caching services



Amazon CloudFront

Static caching (edge caching)

Retrieve content from the nearest edge location.



Amazon ElastiCache

Database caching (data caching)

Retrieve content from an in-memory database layer.

Key takeaways: Overview of caching



- A cache is a high-speed data storage layer that stores a subset of data so that future requests for that data are served up faster.
- Static and frequently accessed data are good candidates for caching.
- Caching improves application speed and decreases database cost.
- Content is retrieved from the nearest edge location with static caching and retrieved from an in-memory database layer with database caching.



Caching using CloudFront

Caching Content

Content delivery network (CDN)

Delivery challenge

Deliver geographically dispersed content with low latency and low cost for end users to access.

CDN solution

- Is a globally distributed system of caching servers
- Has intermediary servers between the client and the application
- Caches copies of commonly requested files (static content)
- Delivers a local copy of the requested content from a nearby cache edge or point of presence

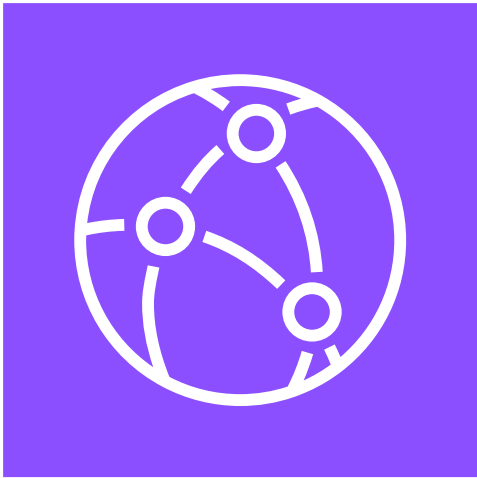
Which type of content can you cache by using an edge cache?

The diagram illustrates the types of content that can and cannot be cached by an edge cache using a screenshot of the Amazon homepage. Annotations include:

- Dynamically generated content cannot be cached by using an edge cache.** (Points to the URL bar and the "Chris's Amazon.com" link)
- User-generated data cannot be cached by using an edge cache.** (Points to the search bar and the "Go" button)
- Web objects can be cached.** (Points to the navigation links at the bottom of the page)
- Images can be cached.** (Points to the image of a Kindle Fire HDX tablet)
- Videos can be cached.** (Points to the "Watch it in Action" video player)

The screenshot shows the Amazon homepage with the URL `https://www.amazon.com/gp/css/homepage.html/ref=gno_yam_ya`. The page includes a search bar, navigation links, and a featured section for "Revolutionary on-device tech support" on Kindle Fire HDX tablets.

CloudFront



CloudFront

- Is a CDN service that delivers content across the globe securely with low latency and high transfer speeds
- Provides high-speed content distribution by delivering through edge locations
- Improves application resiliency from distributed denial of service (DDoS) attacks by leveraging services such as AWS Shield and AWS WAF

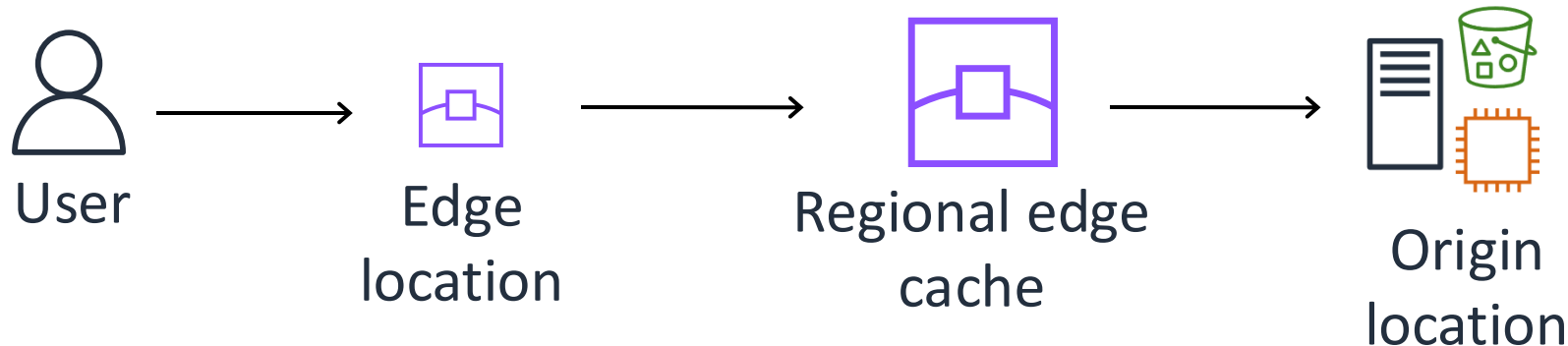
CloudFront global edge network components

CloudFront edge locations

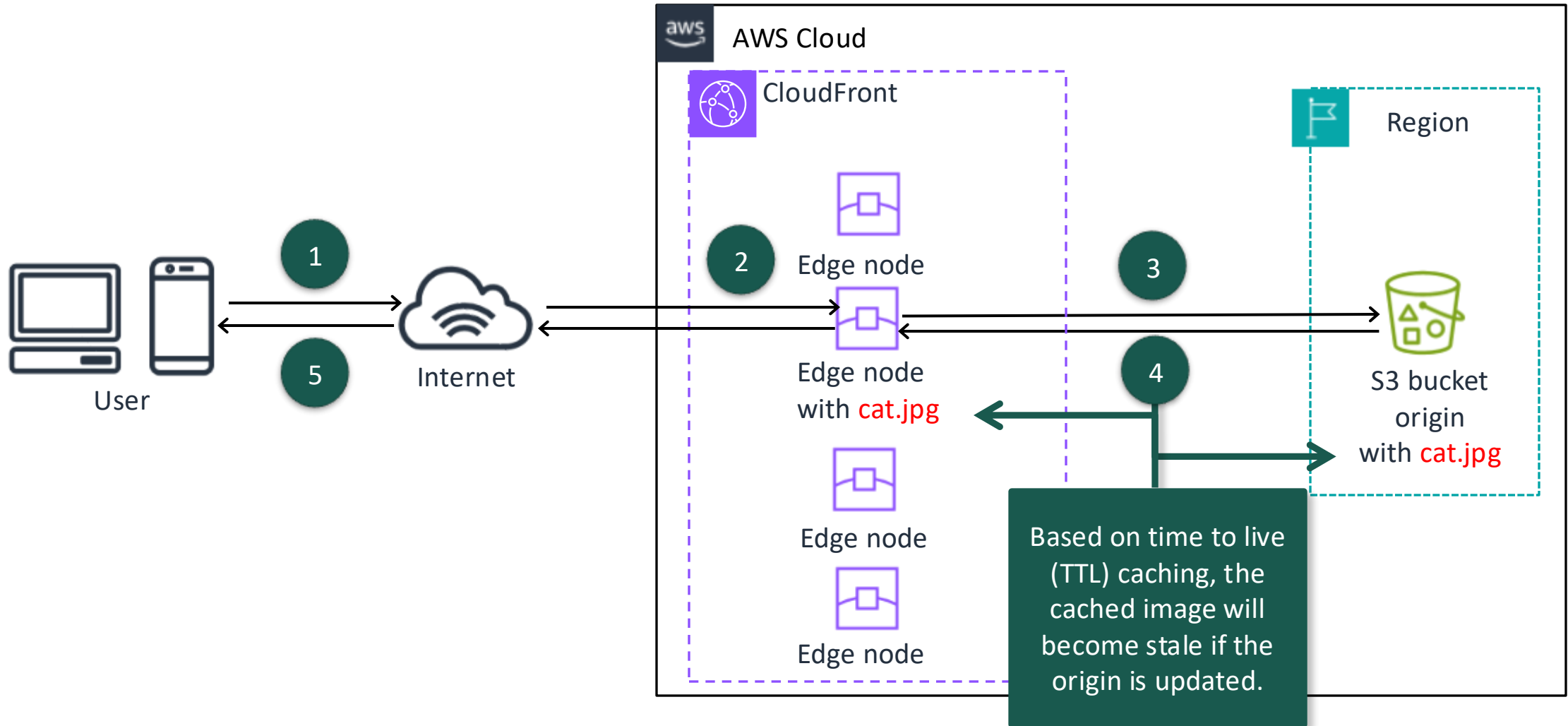
- Are more numerous and closer to users
- Have smaller caches
- Help ensure that popular content can be served quickly to viewers

CloudFront Regional edge caches

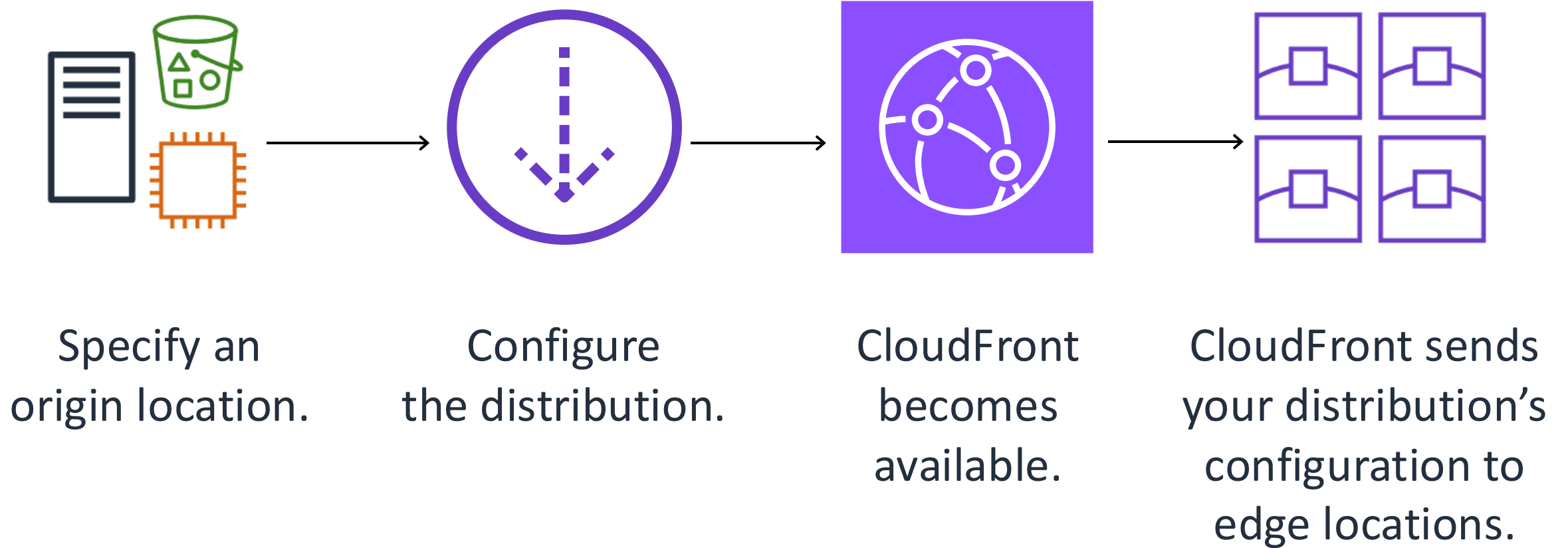
- Are fewer and farther away from users
- Have larger caches
- Helps with less popular content in particular



How caching works in CloudFront



How to configure a CloudFront distribution



Controlling how long content is cached

Set a maximum TTL
value

Expires cached
content faster with a
low maximum TTL
value

Implement content
versioning

Fetches and caches
new files
immediately

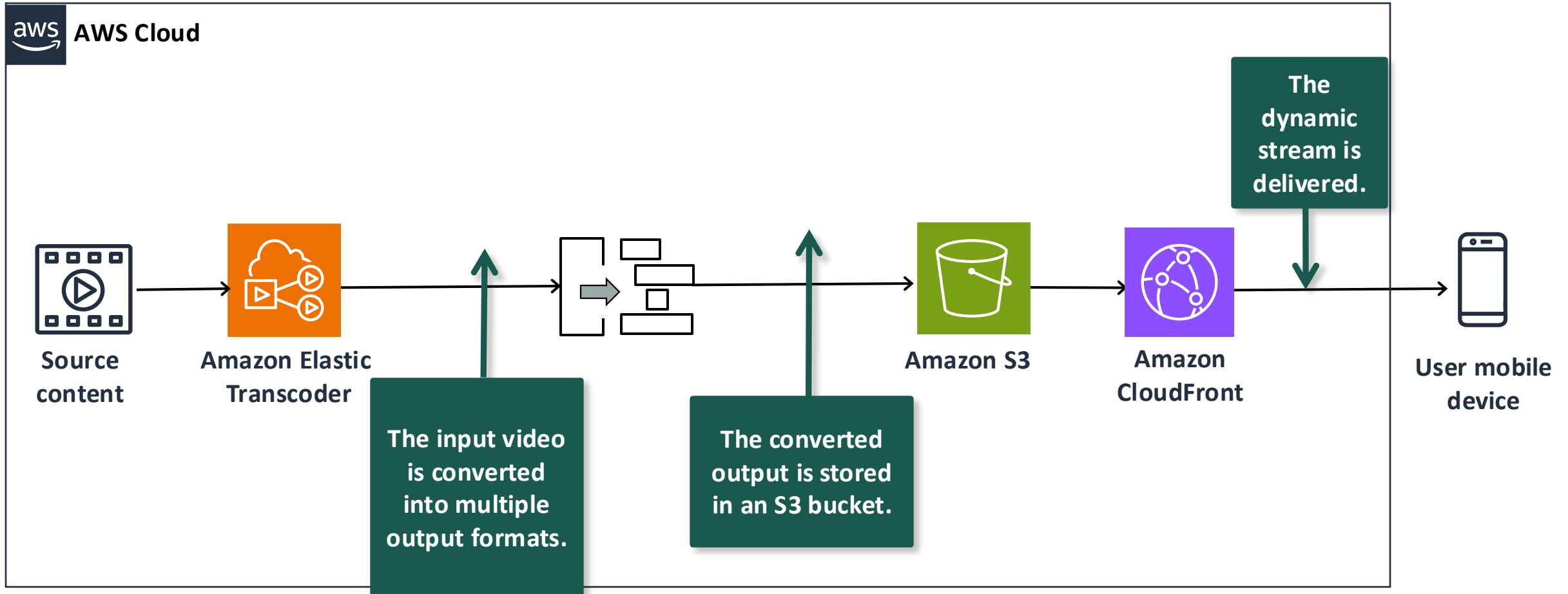
Specify
cache-control
headers

Provides precise
control over content
expiration for
individual files

Use CloudFront
invalidation requests

Forces CloudFront to
expire content from
edge caches

Using CloudFront for video streaming



DDoS mitigation

Security challenge

Publicly accessible content is exposed to common web threats that affect availability and compromise security.

CloudFront solution

- Monitoring and mitigation are built into Amazon Route 53 as it routes traffic to CloudFront.
- CloudFront can create an AWS WAF web access control list (ACL) that configures rules to analyze incoming requests and block threats before they reach your servers.
- Shield DDoS mitigations allow only traffic that is valid for web applications to pass through to CloudFront.

Key takeaways: Caching using CloudFront



- Static file caching is done by using a CDN.
- CloudFront is a CDN service that uses edge locations and Regional edge caches to deliver content securely across the globe.
- You can control CloudFront caching behavior by using a combination of actions.
- By using CloudFront for distributing content, you are also protecting your systems from DDoS attacks.



Guided lab: Streaming Dynamic Content Using Amazon CloudFront (CloudFront lab)

CloudFront lab tasks:



In this lab, you perform the following main tasks:

- Create an Amazon CloudFront distribution.
- Create an Amazon Elastic Transcoder pipeline.
- Test playback of the dynamic (multiple bitrate) stream.

Debrief: CloudFront lab

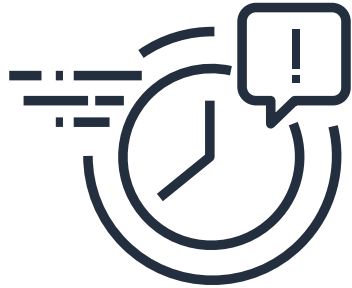
- What did you specify as the origin that will be used to deliver the multiple bit-rate files generated by Amazon Elastic Transcoder to end-user devices?
- Which two components does the playback URL that plays through Amazon CloudFront consist of?



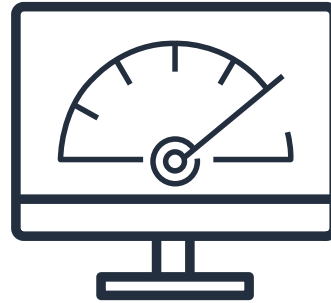
Caching using ElastiCache

Caching Content

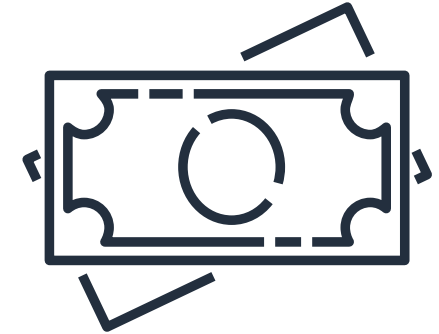
When should you cache your database?



You are concerned about response times for your customer.



You have a high volume of requests that are overburdening your database.



You want to reduce your database costs.

ElastiCache



ElastiCache

- Is a fully managed, key-value, in-memory data store with sub-millisecond latency
- Sits between an application and an origin data store
- Decreases access latency and eases the load of databases and applications
- Provides a high performance and cost-effective in-memory cache
- Is fully compatible with open source Redis and Memcached engines

Choose an ElastiCache engine based on what you need

If you need:

- the most basic model
- the ability to run large nodes with multiple cores or threads
- the ability to scale horizontally with Auto Discovery

Choose



ElastiCache
for Memcached

If you need:

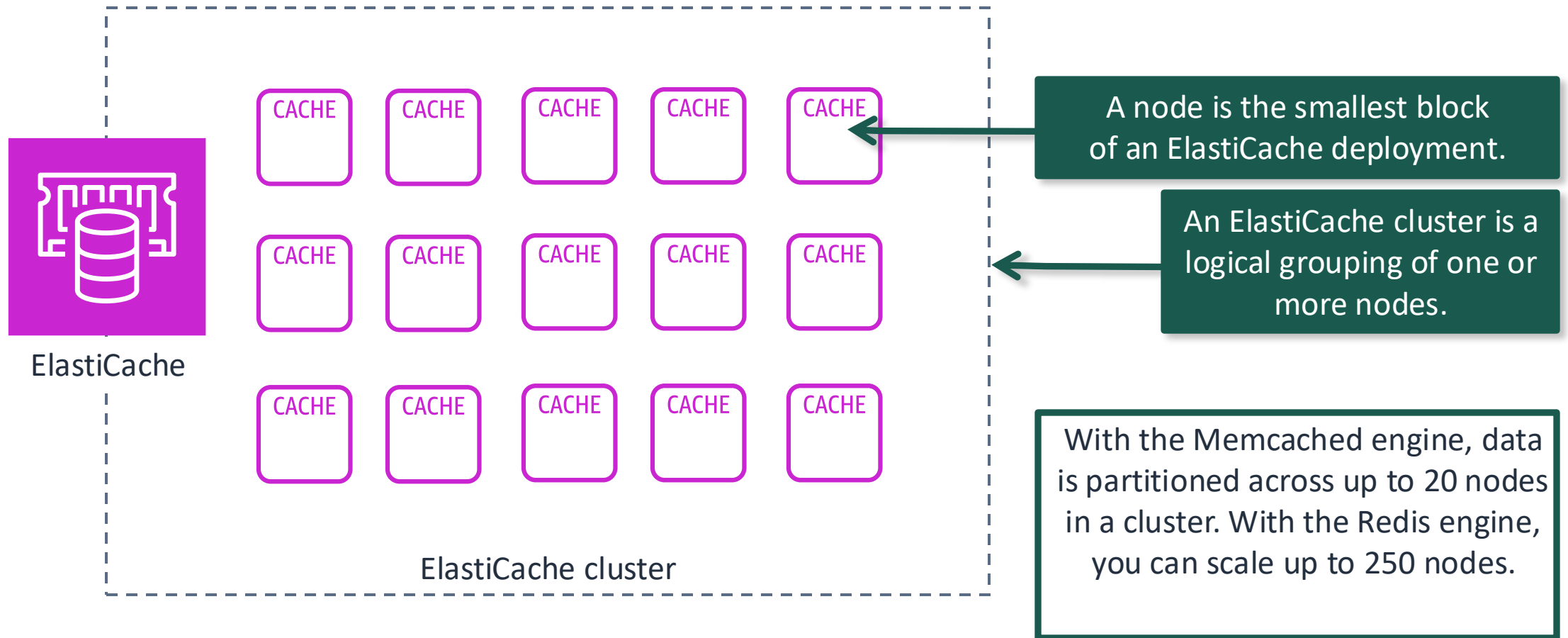
- to support complex data types
- to provide persistence of your key store
- to sort or rank in-memory datasets
- to provide high availability with automatic failover
- to support publish/subscribe messaging

Choose

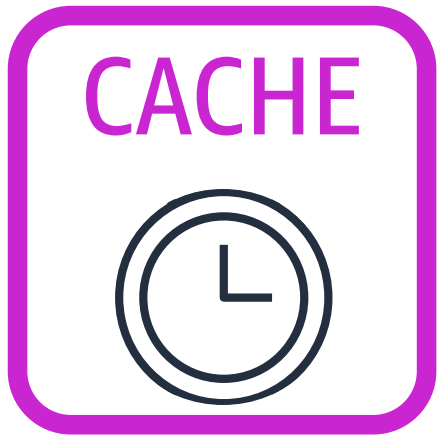


ElastiCache
for Redis

ElastiCache components



Time to live (TTL)

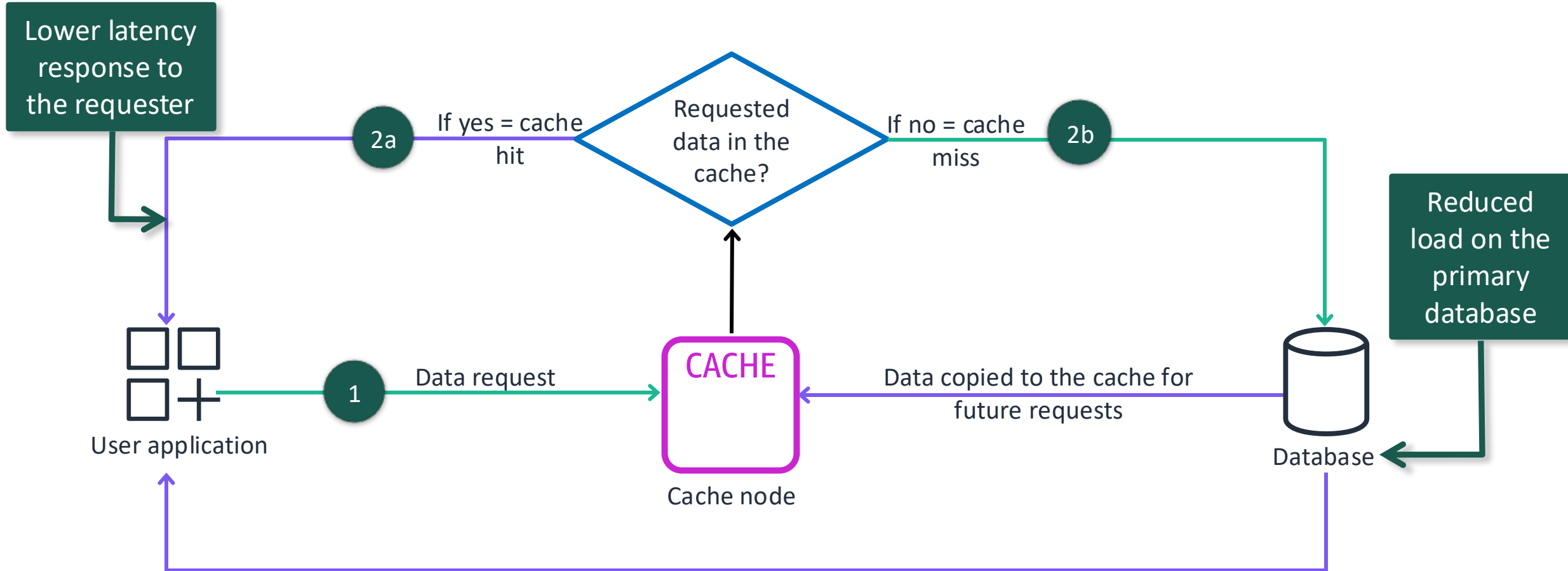


- A TTL value is added to each write.
- The TTL specifies the number of seconds or milliseconds until the key expires.
- After the TTL expires, the application queries the database for data.

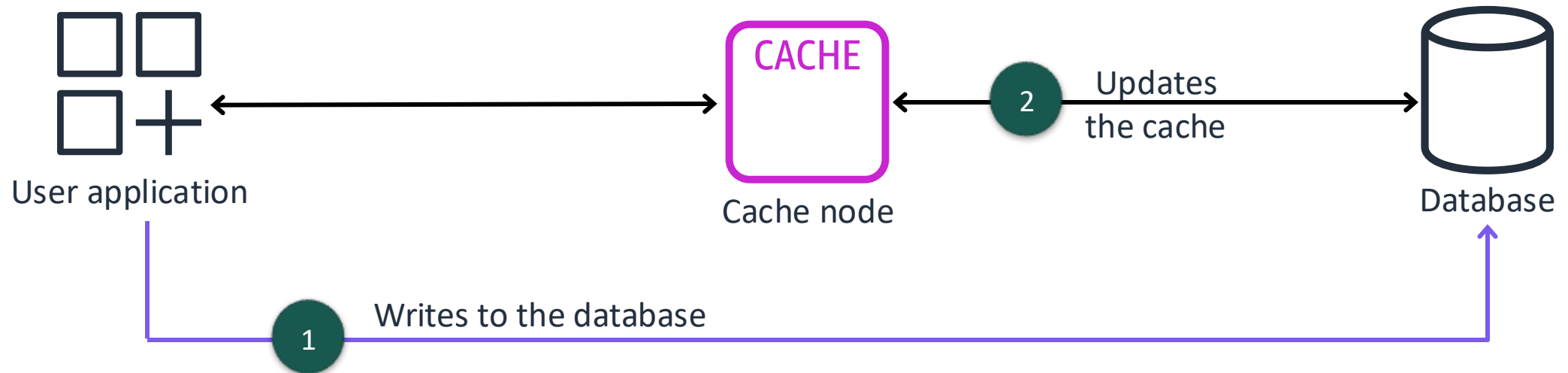
Strategies for handling stale data

Features	Lazy Loading	Write-Through
Caching Pattern	This strategy updates the cache after the data is requested.	This strategy updates the cache immediately after updating the primary database.
Advantage	The cache contains only data that the application actually requests.	The cache is up-to-date with the primary database (most likely data will be found in the cache).
Disadvantage	This strategy requires a programmatic strategy to handle keeping the cache as up to date.	This strategy results in an increase in cost from storing data in-memory that you might not use.

Caching strategies: Lazy loading



Caching strategies: Write-through



Key takeaways: Caching using ElastiCache



- ElastiCache is a key-value, in-memory data store with the purpose of providing millisecond latency and inexpensive access to copies of data.
- ElastiCache reduces the complexity of administering your caching systems for Redis and Memcached engines.
- Lazy loading and write-through are two strategies to handle keeping the cache as up to date as is appropriate.



Applying the AWS Well-Architected Framework principles to caching

Caching Content

AWS Well-Architected Framework caching best practices



Performance
Efficiency



Reliability



Cost
Optimization

Best practice approach: Data management



Performance
Efficiency

Best practices

Implement data access patterns that utilize caching.

Implement strategies to improve query performance in data store.

Best practice approach: Foundations – plan your network topology



Reliability

Best practice

Use highly available network connectivity for your workload public endpoints.

Best practice approach: Cost-effective resources - plan for data transfer



Cost
Optimization

Best practices

Perform data transfer modeling.

Select components to optimize data transfer cost.

Implement services to reduce data transfer costs.

Key takeaways: Applying AWS Well-Architected Framework principles to caching



- Consider best practices in the performance efficiency pillar such as making selection choices based on data characteristics and available options.
- Consider best practices in the reliability pillar such as using highly available network connectivity for your public endpoints.
- Consider best practices in the cost optimization pillar such as implementing services to reduce data transfer costs.



Module wrap-up

Caching Content

Module summary

This module prepared you to do the following:

- Identify how caching content can improve application performance and reduce latency.
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Considerations for the café

- Discuss how you as a cloud architect might advise the café based on the key cloud architect concerns presented at the start of this module.



Module knowledge check



- The knowledge check is delivered online within your course.
- The knowledge check includes 10 questions based on material presented on the slides and in the slide notes.
- You can retake the knowledge check as many times as you like.

Sample exam question

Your company is hosting an ecommerce site and storing the records in an Amazon RDS database. After a few days, the Amazon RDS database is experiencing serious performance issues due to website requests. Which option is the most cost-effective way to deal with the performance issues due to many read requests?

Identify the key words and phrases before continuing.

The following are the key words and phrases:

- Amazon RDS database
- cost-effective
- performance issues due to many read requests

Sample exam question: Response choices

Your company is hosting an ecommerce site and storing the records in an **Amazon RDS database**. After a few days, the Amazon RDS database is experiencing serious performance issues due to website requests. Which option is the most **cost-effective** way to deal with the **performance issues due to many read requests**?

Choice	Response
A	Enable the Multi-AZ feature for the Amazon RDS database.
B	Configure Amazon ElastiCache to cache frequently used content from the database.
C	Place a load balancer in front of the database.
D	Configure Amazon CloudFront to cache content from the database.

Sample exam question: Answer

The answer is B.

Choice	Response
B	Configure Amazon ElastiCache to cache frequently used content from the database.



Thank you

Corrections, feedback, or other questions?

Contact us at <https://support.aws.amazon.com/#/contacts/aws-academy>.